Materials Needed

Dr Donald Frazier, research scientist at NASA's Marshall Space Flight Center at Huntsville, Alabama, says about optical computing, "Entirely optical computers are still some time in the future, but electro-optical hybrids have been possible since 1978, when it was learned that photons can respond to electrons through media such as lithium niobate.

"Newer advances have produced ... thin films and optical fibres that make optical interconnections and devices practical. We are focusing on thin films made of organic molecules ...

"Organics can perform functions such as switching, and signal processing using less power than inorganics. Inorganics, such as silicon used with organic materials, let us use both photons and electrons in current hybrid systems, which will eventually lead to all-optical computer systems."

This underlines the problem of the quest for materials. Dr Frazier, Dr Hossin Abdeldayem, a member of Frazier's optical technologies research group, and the rest of the group, demonstrated all-optical switches, acting as logic gates, operating in the time range of picoseconds.

The Alliance for Non-linear Optics (ANLO), was formed with a grant from NASA with the goal of focusing on materials that support optical computing. Non-linear optics is the key here. In its mission statement, ANLO says that lack of good materials is an impediment to progress, and to that end, experiments with mostly organic materials that may serve as the basis for optical computing devices such as logic gates.

Holographic Storage

The hard disk is probably the most important component of a personal computer. So what about the 'opticalisation' of storage? The answer lies in holographic storage, which uses the volume of a medium, rather than its surface to record and read data. This would result in vastly increased densities and read-write speeds, while being, at the same time, ultra-reliable and cost-effective.

The principle relies on lasers: Light from a laser beam is split into two—a signal beam that carries data, and a reference beam, which is used to write and read to the holograms. A device—the spatial light modulator (SLM)—translates the 0s and 1s of data into an optical two-dimensional array of pixels that are either lit or dark. At the point where the signal beam and the reference beam intersect, a hologram is formed because of a chemical reaction between the light-sensitive recording medium and the lit elements of the signal beam. This hologram is then recorded on the medium.

By varying the angle and wavelength of the reference beam, or the position of the medium, different holograms can be recorded—basically, superimposing holograms within the same space. Reading is accomplished by flashing the reference beam onto the media: The reference beam deflects off the stored hologram. Data is read parallely since the deflected beam is projected onto a detector that reads the aspects of the hologram in parallel.

Again, note that holographic storage is not a new concept: It's been researched for nearly four decades now.

Holographic storage relies on the 'etching' of a hologram onto a medium

Jargon Buster

DWDM: Dense Wavelength Division Multiplexing is a technology that lets you transmit more data over a fibre-optic network than is usually possible. To do so, it sub-divides signals of a particular frequency on a single fibre into discrete wavelengths and multiplexes them. These wavelengths are recognised at the receiving end.

OEO transponder: An OEO transponder converts light signals to electric ones, and then back to light. This is necessary because certain operations on the signals can, as of now, be done only in electronic mode.

Photonic crystals: Structures made of an insulating medium that allows light to pass through them, with a property that forbids the propagation of a certain frequency range. This enables comparatively very easy control of light.

Non-linear optics: The field that deals with the interaction of matter with strong light fields.

Optical waveguide: Used in fibre-optic networks, an optical waveguide is used to control propagation of light from a source.

Core IP router: A high-throughput router that's at the hub of several interconnected networks.

The optical computer

There's no race on to build a fully-optical computer. Rather, each of the advancements that we have just mentioned does a particular job, or demonstrates a particular principle. The real race, if any, is that of finding materials that do the job well—for example, materials that are capable of storing holograms in holographic storage, or those ideally suited for building optical transistors.

The question of why we're not at an optical computer yet might arise: With optical processors, optical data paths, etc, demonstrated, why not build an optical computer right now? The answer is that these components—for example, the optical processor by Lenslet—are isolated; none of them are designed to work with each other. There's no consortium that has defined any kinds of standard for 'optical processing' or 'optical computing'. Research is scattered, and